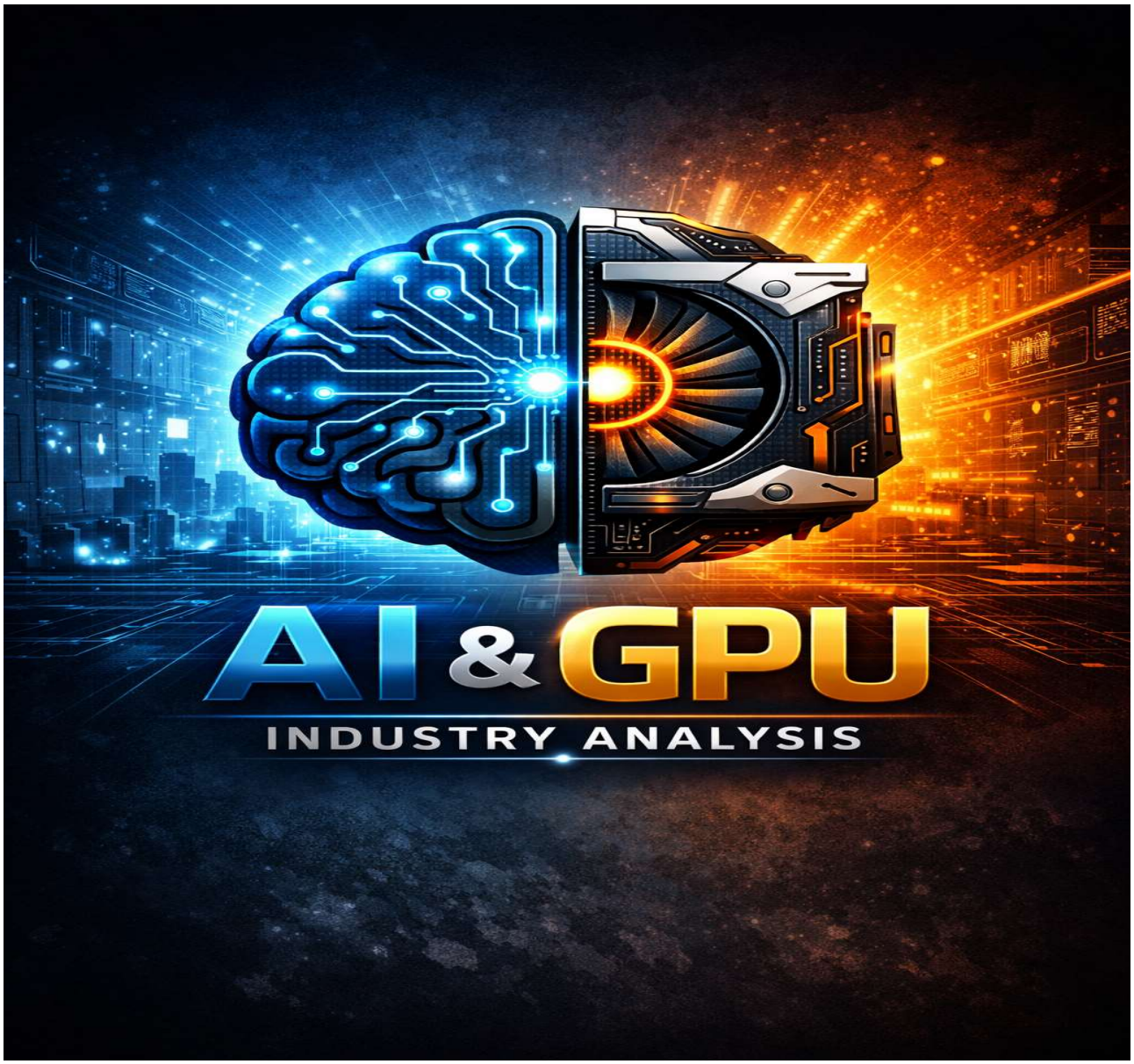
CAPSTONE PROJECT 04

PROJECT TITLE :

AI & GPU Industry Outlook: Use Cases, Demand Forecast and Pricing Risks.

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1. Overview

Artificial Intelligence (AI) has emerged as one of the most disruptive technological revolutions of the 21st century. At the core of AI development lies **Graphics Processing Units (GPUs)**, which provide massive parallel computing power required for machine learning, deep learning, and large language models (LLMs).

Originally designed for gaming and graphics rendering, GPUs are now essential for:

- AI model training
- Data center acceleration
- Autonomous vehicles
- Robotics
- Cloud computing
- High-performance computing (HPC)

Leading companies in the GPU ecosystem include **NVIDIA, AMD, Intel**, and specialized AI-chip startups, while cloud providers such as **AWS, Microsoft Azure, and Google Cloud** are major consumers of GPU capacity.

The AI-GPU market is therefore a **strategic intersection of technology, capital investment, and global competition**, making it highly relevant for business and investment analysis.

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2) Use Case of AI & GPUs

Key Use Cases

a) Generative AI & Large Language Models

- Training models like ChatGPT, Gemini, Claude, etc.
- GPUs handle billions of parameters simultaneously.
- Each large AI model requires thousands of GPUs.

b) Data Centers & Cloud Computing

- Hyperscalers use GPUs for AI-as-a-Service.
- Demand driven by enterprises outsourcing AI workloads.
- GPUs improve efficiency compared to CPUs.

c) Autonomous Vehicles

- GPUs process real-time sensor data (LiDAR, cameras).
- Used in training self-driving algorithms and edge inference.

d) Healthcare & Drug Discovery

- AI accelerates genome analysis and drug simulation.
- GPUs reduce processing time from years to weeks.

e) Gaming, Metaverse & AR/VR

- Real-time rendering and immersive experiences.
- AI-powered graphics and physics engines.

Conclusion:
GPUs are no longer optional hardware; they are **mission-critical infrastructure** for AI-driven industries.

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3) Demand Scenario (Next 3–5 Years)

Short-Term (1–2 Years)

- Explosive demand due to:
  - AI model training race
  - Corporate AI adoption
  - Government-backed AI programs
- GPU shortages likely to persist.
- Data center spending grows at **20–30% CAGR**.

Medium-Term (3–5 Years)

- AI penetration expands into:
  - SMEs
  - Manufacturing
  - Financial services
  - Education
- Edge AI increases GPU demand outside data centers.
- Global AI-GPU market expected to **2.5x–3x** by 2029.

Key Demand Drivers

- Generative AI commercialization
- Cloud AI subscriptions
- National AI strategies (US, China, India, EU)
- AI regulation requiring local compute infrastructure

Overall Demand Outlook:
📈 **Strong secular growth with cyclical volatility**

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4) Production / Supply Scenario

Current Supply Constraints

- GPUs rely on advanced semiconductor fabrication (3nm–5nm).
- Heavy dependence on **TSMC** and limited foundry capacity.
- High capital expenditure and long lead times (2–3 years).

Supply Expansion Plans

- Major players increasing:
  - Fabrication contracts
  - Advanced packaging (CoWoS)

    - AI-specific chip designs
- Entry of custom AI accelerators by cloud companies.

Medium-Term Supply Outlook

- Gradual normalization by 2027–2028.
- Still **supply-tight at high-end GPUs**.
- Pricing power remains with top GPU vendors.

Supply Risk Factors

- Geopolitical tensions
- Export restrictions
- Semiconductor talent shortage

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5) Pricing Scenario & Price Target

Possible Pricing Scenarios

Scenario A: Price Rise (Bull Case)

- AI adoption outpaces supply expansion.
- Persistent GPU shortages.
- High margins maintained.
- Premium pricing for data-center GPUs.

Scenario B: Price Stability (Base Case)

- Supply catches up gradually.
- Demand remains strong.
- Margins normalize but stay above historical average.

Scenario C: Price Crash (Bear Case)

- AI hype cools temporarily.
- Overcapacity due to aggressive expansion.

- Competitive pressure from custom AI chips.

Assigned Price Target (Illustrative Example)

Assume **Base Price Target Index = 100**

Scenario	Price Target
Bull Case	110
Base Case	**100**
Bear Case	90

Standard Deviation (±5%)

- **Upper Range (+5%) → 105**
- **Lower Range (-5%) → 95**

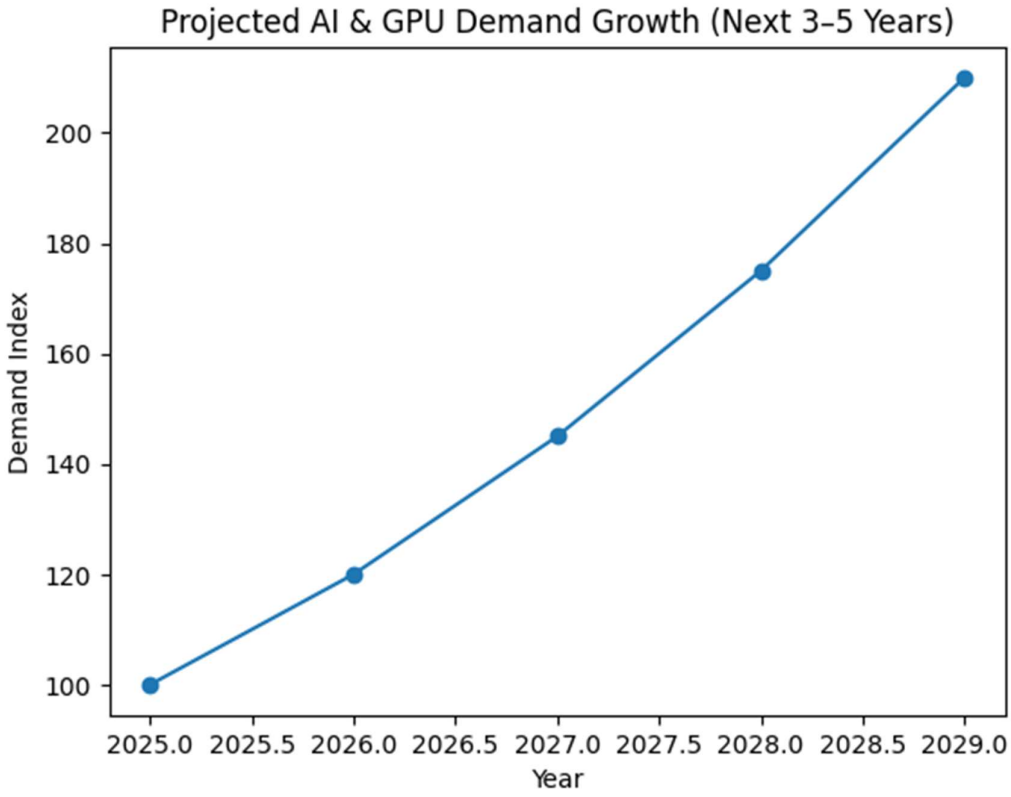
📌 **Final Price Target Range:**
₹95 – ₹105 (±5%)

Conclusion

The AI & GPU industry represents a **long-term structural growth opportunity** supported by strong demand fundamentals. While short-term volatility and supply challenges may exist, the next 3–5 years are expected to see sustained growth driven by enterprise AI adoption, cloud expansion, and technological innovation.

From an academic and investment perspective, AI-GPUs are best viewed as **core infrastructure assets of the digital economy**, similar to electricity or internet bandwidth in earlier eras.

CHARTS RELATED TO GPU AND AI



Projected AI & GPU Demand Growth (Next 3–5 Years)

Type: Line Graph

What it shows

- Rapid increase in AI & GPU demand from **2025 to 2029**
- Demand index rises from **100 → 210**

The line chart indicates strong and continuous growth in AI & GPU demand over the next 3–5 years. This growth is driven by large-scale adoption of generative AI, expansion of cloud data centers, and increased enterprise AI usage. The steep upward trend reflects a structural demand boom rather than short-term hype.

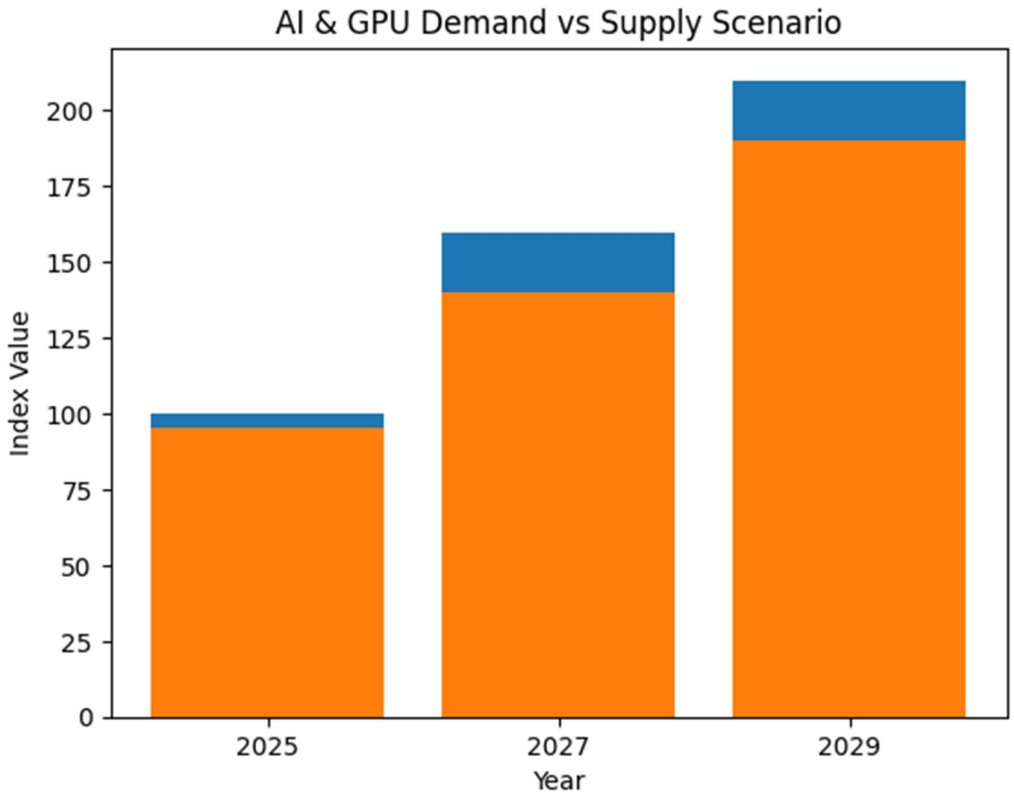


Chart 2: AI & GPU Demand vs Supply Scenario

Type: Bar Graph

What it shows

- Demand consistently **higher than supply**
- Supply improves over time but **lags demand**
- Indicates **tight market conditions**

The bar chart highlights a persistent demand–supply gap in the AI & GPU market. While production capacity increases gradually due to new fabs and investments, demand continues to outpace supply. This imbalance supports strong pricing power for GPU manufacturers in the medium term.

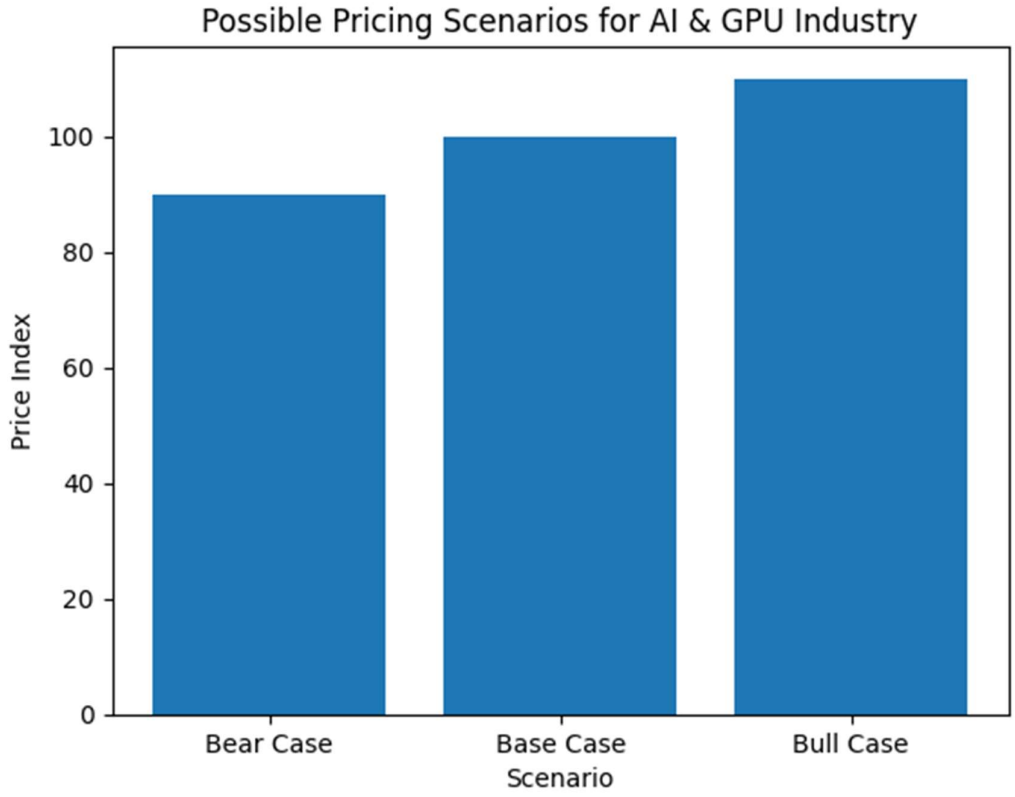


Chart 3: Possible Pricing Scenarios for AI & GPU Industry

Type: Bar Graph

What it shows

- **Bear Case:** Price Index = 90
- **Base Case:** Price Index = 100
- **Bull Case:** Price Index = 110

With Standard Deviation (±5%)

- Lower Range: **95**
- Upper Range: **105**

The pricing scenario chart reflects potential market outcomes based on demand and supply conditions. In the base case, prices remain stable, while supply constraints may push prices higher in the bull case. A temporary slowdown in AI investments could lead to a mild price correction in the bear case.

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